Trigonometry: Radians, Arcs and Sectors

Questions 1–25

Question 1. Copy and complete the following table:

Degrees	Radians
180°	
90°	
	$\frac{\pi}{3}$
	$\frac{\pi}{4}$
	$\frac{\pi}{5}$
30°	
20°	
	$\frac{8\pi}{9}$
1.62°	
	1.62

Question 2. Find the length of each of these sector's arc with angle θ radians and radius r:

a) $\theta = 0.5$, $r = 10$ cm
b) $\theta = \frac{\pi}{4}$, $r = 20$ m
c) $\theta = 0.75$, $r = 15$ cm
d) $\theta = \frac{3\pi}{8}$, $r = 120$ m

Question 3. Find the area of each of these sectors with angle θ radians and radius r:

a) $\theta = 0.4$, $r = 20$ cm

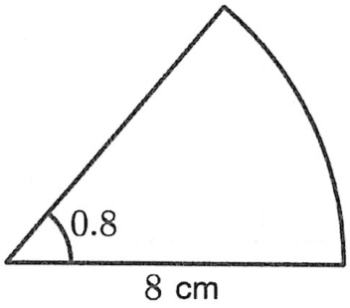
b) $\theta = \frac{\pi}{3}$, $r = 9$ m

c) $\theta = 1.15$, $r = 64$ cm

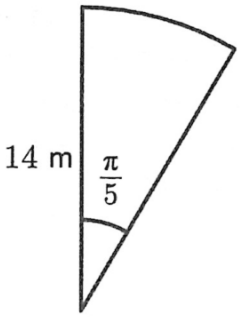
d) $\theta = \frac{\pi}{20}$, $r = 18.5$ m

Question 4. Find the area and perimeter of the following sectors, given that the angle shown is measured in radians:

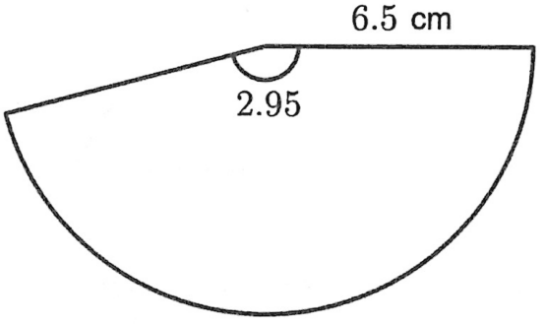
a)



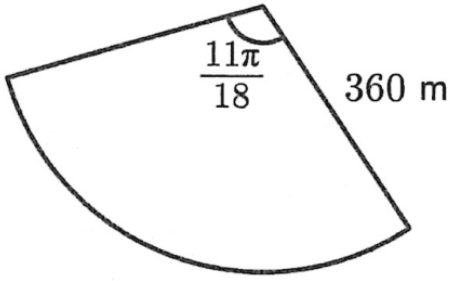
b)



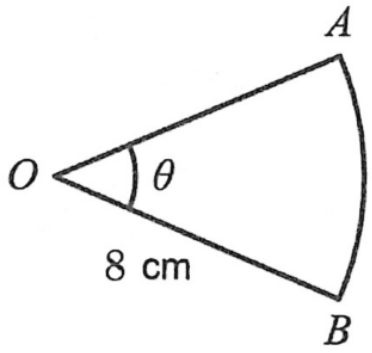
c)



d)



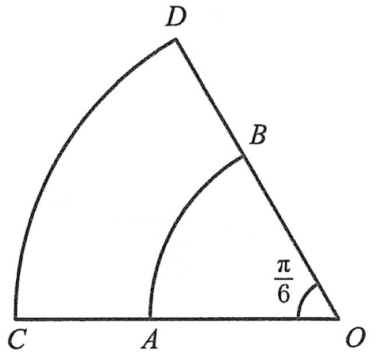
Question 5. A sector of a circle, OAB, with radius 8 cm has perimeter 25 cm. The angle AOB is θ radians. Find the value of θ.



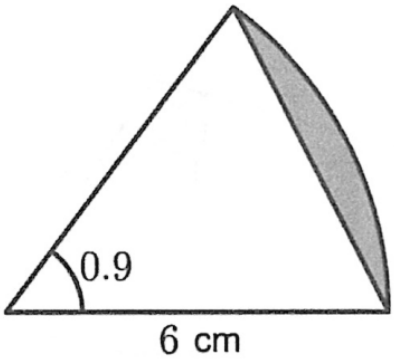
Question 6. A sector of a circle with radius r cm and angle θ radians has an area of 50 cm^2. Write down an equation for r in terms of θ

Question 7. A sector of a circle with radius r cm and angle θ radians has a perimeter of 84 m. Write down an equation for θ in terms of r

Question 8. The diagram below shows two sectors, OAB and OCD. OAC and OBD are straight lines. The angle AOB is $\frac{\pi}{6}$ radians. The length of OC is 9 cm. It is given that the area of sector OCD is three times larger than the area of OAB. Find the exact length of OA.

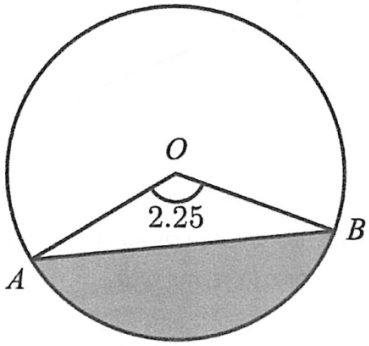


Question 9. The diagram below shows a sector of a circle with radius 6 cm and angle 0.9 radians.



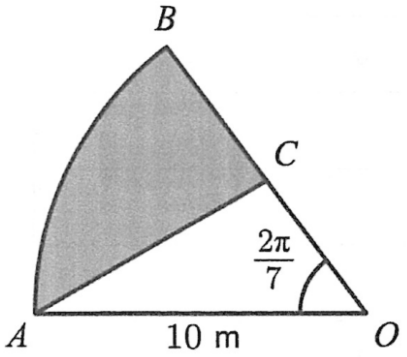
a) Find the area of the shaded segment.
b) Find the perimeter of the shaded segment.

Question 10. The diagram below shows the circle with equation $x^2 + y^2 - 4x + 14y + 19 = 0$. It is given that OAB is a sector with angle 2.25 radians.

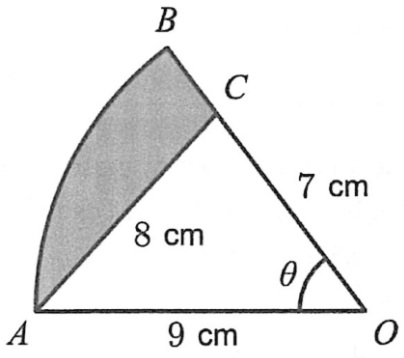


a) Find the area of the shaded segment.
b) What percentage of the circle is shaded in the diagram?

Question 11. The diagram below shows a sector, OAB, of a circle with radius 10 m and angle $\frac{2\pi}{7}$ radians. It is given that C is the midpoint of O and B. Find the shaded area.



Question 12. The diagram below shows a sector, OAB, of a circle with radius 9 cm and angle θ radians. It is given that AC has length 8 cm, and that OC has length 7 cm.

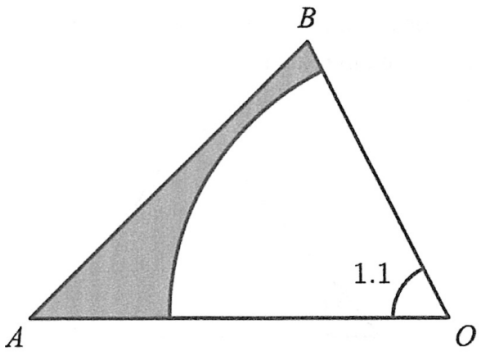


a) Show that $\cos\theta = \frac{11}{21}$

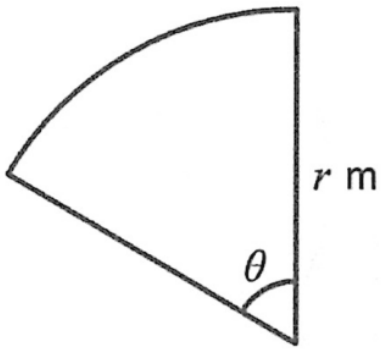
b) Hence show that $\sin\theta = \frac{8\sqrt{5}}{21}$

c) Find the area of the shaded region.

Question 13. The diagram below shows the triangle OAB. Within the triangle is a sector of a circle, where angle AOB is 1.1 radians. It is given that $OA = 15$ cm and $OB = 5\sqrt{5}$ cm. The area of the shaded region is 20 cm^2. Find the perimeter of the shaded region to 3 significant figures.

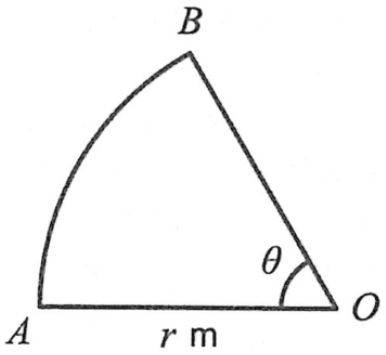


Question 14. The diagram below shows a sector of a circle with radius r m and angle θ radians. It is given that the area of the sector is 2 m^2 and the perimeter is 6 m.

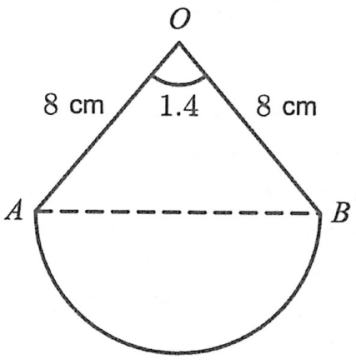


a) Show that $r^2 - 3r + 2 = 0$
b) Hence find the two possible values of θ

Question 15. The diagram below shows a sector of a circle with radius r m and angle θ radians. It is given that the area of the sector is 45 m^2 and the perimeter is six times the length of the arc AB. Find the values of r and θ

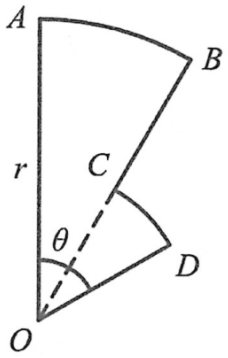


Question 16. The diagram below shows the cross-section of a design for a new spinning toy.

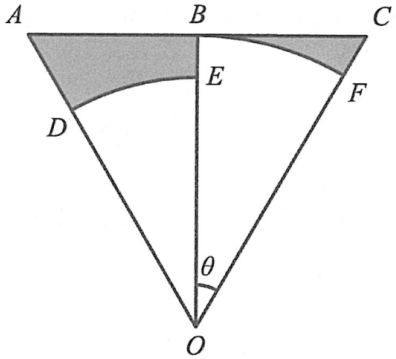


The shape is constructed from an isosceles triangle OAB and a semi-circle with diameter AB. The toy is redesigned so that the cross-section is instead a sector of a circle with the same area as the original design. The sector has an angle of 1.4 radians and radius r cm. Determine the value of r correct to 3 significant figures.

Question 17. A company logo is made from two similar sectors, as shown in the diagram below, where OA is twice the length of OD. The length of OA is r and the angle AOB is θ radians. It is given that the perimeter and area of the whole logo is 9.2 and 2 respectively. Show that there is only one valid pair of values for r and θ

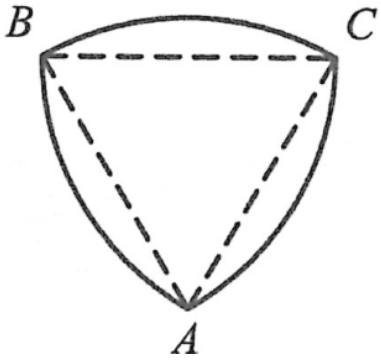


Question 18. It is given that OAC is an equilateral triangle with side length r, and the angle BOC is θ radians. It is also given that OBF and ODE are sectors. B is the midpoint of AC.

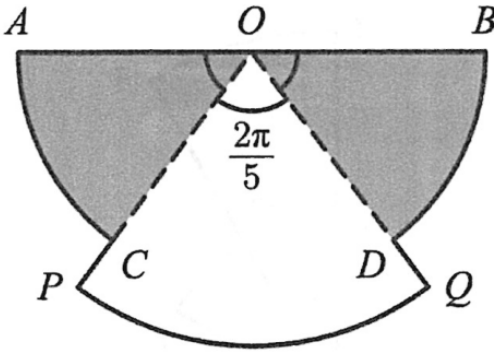


Given that $OB = 5BE$, show that the shaded area can be written in the form $(a\sqrt{3} + b\pi)r^2$, where a and b are rational numbers.

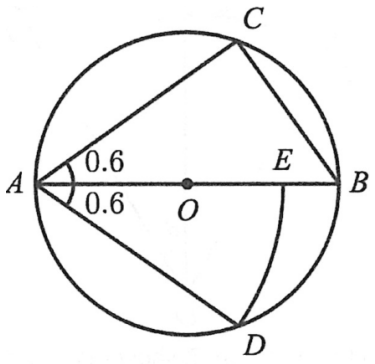
Question 19. The diagram below shows an equilateral triangle, ABC, with side length 5 cm. Glued to the edge of each side of the triangle is the segment of a circle created at the opposite corner. Find the total exact area of the badge.



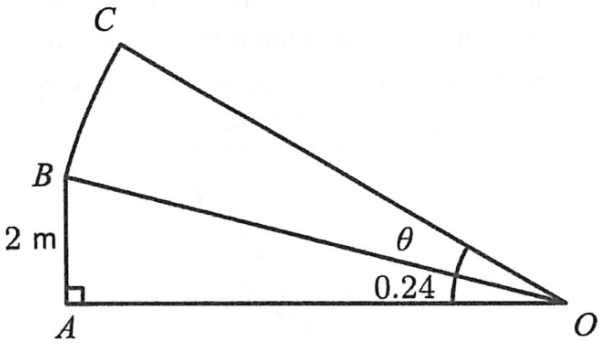
Question 20. The diagram below shows part of a design for a playground. It is made from three sectors. The length of the arc PQ is 12π m. The two sectors OAC and OBD are precisely the same size, and their combined area is equal to the area of the sector OPQ. It is given that AOB is a straight line. Find the length of AB.



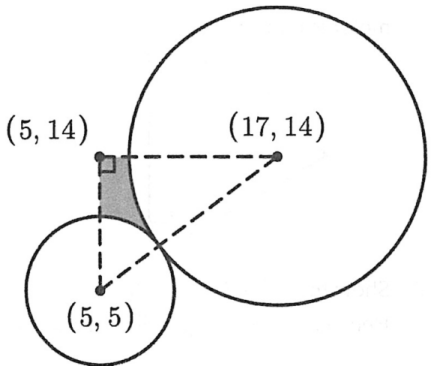
Question 21. The diagram below shows the circle with equation $x^2 + y^2 - 6x + 12y - 4 = 0$. It is given that AB is a diameter of the circle, and both C and D lie on the circle. The angle CAB and the angle DAE are both equal to 0.6 radians. ADE is a sector of a circle centred at A. Find the area of the sector ADE to 3 significant figures.



Question 22. In the diagram below, OAB is a right-angled triangle and OBC is a sector of a circle centred at O. Triangle OAB and sector OBC have the same area. The angle BOC is θ radians. Find the shortest distance from C to the line AO

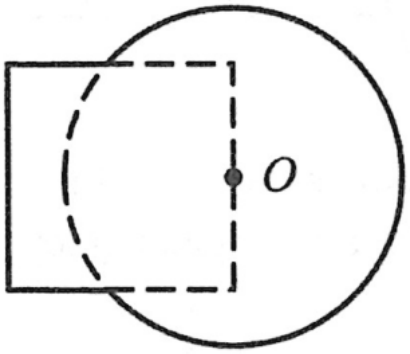


Question 23. The diagram below shows the two circles with equations $(x-5)^2+(y-5)^2=25$ and $(x-17)^2+(y-14)^2=100$



a) Show that the circles intersect at a single point.
b) Find the shaded area.

Question 24. A square is placed over the top of a circle, as shown in the diagram below, to create a new shape. The circle, centred at O, has equation $x^2+y^2=36$. The point O is the midpoint of one of the sides of the square. The square has an area of 64 units^2. Find the perimeter of the new shape, ignoring any dashed lines.



Question 25. The diagram below shows the two circles $x^2 + y^2 = 16$ and $(x - k)^2 + y^2 = 16$. It is given that the perimeter of the shape formed by the overlapping circles is 10π. Find the exact value of k, given that $k > 0$

